

Name: Anthony Claudet

Date: 5/3/25

Class: Programming for Engineers

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1 (HIGH Priority)

The program must evaluate each device and recommend delaying, turning off, or keeping the device on based on whether it is peak energy hours and whether the device is essential.

This directly addresses the failures found in manual scheduling and static timers by allowing dynamic, real-time decision-making based on device essentiality and energy timing.

The design requirement was **approved** by the stakeholder.

Design Requirement #2 (HIGH Priority)

The program must allow user input to define each device's name, power usage (in kW), and whether it is essential or non-essential.

Prior designs lacked flexibility. Manual entry ensures the system can adapt to each household's unique setup instead of using fixed assumptions.



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The design requirement was **approved** by the stakeholder.

Design Requirement #3 (HIGH Priority)

The program must categorize time of day as either "peak" or "off-peak" hours based on user input and adjust recommendations accordingly.

Static systems didn't account for real-time energy demand. Categorizing by time allows the simulation to model how real grids experience peak pricing and congestion.

The design requirement was **approved** by the stakeholder.

Design Requirement #4 (HIGH Priority)

The program must output recommendations for each device in plain, understandable language.

Green City Times emphasized that users often don't act unless recommendations are easy to understand. Clear output helps users make better decisions.

The design requirement was **approved** by the stakeholder.



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Design Requirement #5 (HIGH Priority)

The program must be able to analyze at least ten devices in one session without crashing or freezing.

Smart homes typically have multiple connected devices (ScienceDirect), and real systems must manage many devices. Handling multiple entries tests the system's stability and basic scaling ability. This was previously set to five devices, but has been doubled, requiring it to handle a more realistic number.

The design requirement was **approved** by the stakeholder.

Design Requirement #6 (LOW Priority)

The program should allow users to edit device information (such as changing whether a device is essential) after initial input.

Not critical for a basic simulation but reflects flexibility issues highlighted in earlier static designs (ResearchGate). Useful for improving user experience later.

The design requirement was **approved** by the stakeholder.



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Design Requirement #7 (LOW Priority)

The program should provide a summary report showing total recommended energy savings based on delayed devices.

Environmental reports (ScienceDirect) emphasize the importance of quantifying savings, but a summary report isn't essential for the basic decision-making simulation.

The design requirement was **approved** by the stakeholder.